Earners AI and ML by Aditya Jain sir -

Summary Notes

Lecture 13 Recap - Dictionary

```
student = {
    "name" : "Raj",
    "marks" : 95,
    "city" : "Mumbai"
}
```

```
print(student)
```

```
{'name': 'Raj', 'marks': 95, 'city': 'Mumbai'}
```

```
type(student)
```

```
dict
```

```
student["name"]
```

```
'Raj'
```

```
student["marks"]
```

```
95
```

```
student["city"]
```

```
'Mumbai'
```

```
student[0]
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipykernel_1082/3657199547.py in <cell line: 0>()
----> 1 student[0]

KeyError: 0
```

Next steps: Explain error

```
student.get(0)
```

```
student.get("city")
```

```
'Mumbai'
```

```
student.get("state", "key not found")
```

```
'key not found'
```

```
student.get("city", "key not found")
```

```
'Mumbai'
```

```
for k in student:
  print(k)
```

```
name
marks
city
```

```
for k in student.keys():
  print(k)
```

```
name
marks
city
```

```
for v in student.values():
  print(v)
```

```
Raj
95
Mumbai
```

```
for k in student.keys():
  print(student[k])
```

```
Raj
95
Mumbai
```

```
for item in student.items():
  print(item)
```

```
('name', 'Raj')
('marks', 95)
('city', 'Mumbai')
```

HW

```
fruit = {
    "Apple": 50,
    "Banana": 20
}

fruit["Apple"] += fruit["Banana"]

print(fruit["Apple"])
print(fruit["Banana"])
```

```
70
20
```

```
student = {
    "Math": 90,
    "Science": 85,
    "English": 95
}

highest = 0

for subject in student:
    if student[subject] > highest:
        highest = student[subject]

print(highest)
```

```
95
```

```
student = {
    "name": "Rahul",
    "age": 22
}

student["age"] = student["age"] + len(student)

print(student["age"])
```

```
24
```

```
data = {}

data["A"] = 10
```

```
data["B"] = 20
data["A"] = 30

print(len(data))
```

```
data = {}
```

```
type(data)
```

```
dict
```

```
len(data)
```

```
0
```

```
data = {}

data["A"] = 10
data["B"] = 20
data["A"] = 30

print(len(data))
```

```
2
```

```
student = {
    "Math": 80,
    "Science": 90,
    "English": 85
}

count = 0

for key in student:
    if student[key] >= 85:
        count += 1

print(count)
```

```
2
```

```
person = {
    "name": "Amit",
    "age": 25
}

person.pop("age")

person["age"] = 30

print(person)
```

```
{'name': 'Amit', 'age': 30}
```

```
marks = {
    "Math": 90,
    "Science": 80,
    "English": 70
}

total = 0

for subject in marks:
    if marks[subject] >= 80:
        total += marks[subject]

print(total)
```

```
170
```

```
student = {
    "Math": 95,
    "Science": 85
```

```
}

for key in student:
    print(key, student[key] // 10)
```

```
Math 9
Science 8
```

```
95//10
```

```
9
```

```
95/10
```

```
9.5
```

```
85/10
```

```
8.5
```

```
85//10
```

```
8
```

```
data = {
    "A": 5,
    "B": 10,
    "C": 15
}

result = 0

for key in data:
    if data[key] % 10 == 0:
        result += 1

print(result)
```

```
1
```

```
student = {
    "Math": 90,
    "Science": 80
}

student["Total"] = 0

for key in student:
    if key != "Total":
        student["Total"] += student[key]

print(student["Total"])
```

```
170
```

Tuple

```
days = ["Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"]
days[0] = "AJ Day"
```

```
days
```

```
['AJ Day', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', 'Saturday', 'Sunday']
```

```
days = ("Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday")
```

```
days[0] = "AJ Day"

#Tuples are immutable
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1082/1721256602.py in <cell line: 0>()
----> 1 days[0] = "AJ Day"

TypeError: 'tuple' object does not support item assignment
```

Next steps: Explain error

```
a = ("AJ", 9.9, 24, True)
type(a)
```

```
tuple
```

```
a[2]
```

```
24
```

```
b = ()
len(b)
```

```
0
```

```
type(b)
```

```
tuple
```

```
c = (1)
type(c)
```

```
int
```

```
(3+5)
```

```
8
```

```
d = (1,)
```

```
type(d)
```

```
tuple
```

```
len(d)
```

```
1
```

```
a
```

```
('AJ', 9.9, 24, True)
```

```
print(a[1])
```

```
9.9
```

```
print(a[-1])
```

```
True
```

```
print(a[-2])
```

```
24
```

Traversing the Tuples

```
for elem in a:
  print(elem)
```

```
AJ
9.9
24
True
```

```
n = len(a)

for i in range(n):
  print(a[i])
```

```
AJ
9.9
24
True
```

```
a
```

```
('AJ', 9.9, 24, True)
```

```
print(a[1:3])
```

```
(9.9, 24)
```

```
b = ("p", "q", "r")
print("p" in b)
```

```
True
```

```
b = ("p", "q", "r")
print("a" in b)
```

```
False
```

```
b = ("p", "q", "r")
print("a" not in b)
```

```
True
```

```
num = (2,4,5)
print(sum(num))
```

```
11
```

```
num = (2,4,5,4)
print(num.count(4))
```

```
2
```

```
num.index(4)
```

```
1
```

Packing and Unpacking

```
student = ("Raj", 9.9, 24)
```

```
type(student)
```

```
tuple
```

```
#Unpacking
name, cgpa, age = student
```

```
print(name)
```

```
Raj
```

```
print(cgpa)
```

```
9.9
```

```
print(age)
```

```
24
```

```
name, cgpa, age = ("Raj", 9.9, 24)
print(name, age, cgpa)
```

```
Raj 24 9.9
```

```
def show():
  return 2,3
```

```
a = show()
print(a)
```

```
(2, 3)
```

Quiz

```
numbers = (10, 20, 30, 40)

print(numbers[2])
```

```
30
```

```
fruits = ("Apple", "Banana", "Mango")

print(fruits[-2])
```

```
Banana
```

```
t = (5)

type(t)
```

```
int
```

```
t = (5,)

type(t)
```

```
tuple
```

```
numbers = (10, 20, 30)

print(len(numbers))
```

```
3
```

```
numbers = (15, 25, 5, 40)

print(max(numbers))
```

```
40
```

```
numbers = (15, 25, 5, 40)

print(min(numbers))
```

```
5
```

```
marks = (80, 85, 90)

print(sum(marks))
```

```
255
```

```
numbers = (1, 2, 2, 3, 2)

print(numbers.count(2))
```

```
3
```

```
fruits = ("Apple", "Banana", "Mango", "Orange")

print(fruits.index("Mango"))
```

```
2
```

```
colors = ("Red", "Green", "Blue")

for color in colors:
    print(color)
```

```
Red
Green
Blue
```

```
colors = ("Red", "Green", "Blue")

for elem in colors:
    print(elem)
```

```
Red
Green
Blue
```

```
fruits = ("Apple", "Banana", "Mango")

print("Banana" in fruits)
```

```
True
```

```
fruits = ("Apple", "Banana", "Mango")

print("banana" in fruits)
```

```
False
```

```
numbers = (10, 20, 30)

numbers[1] = 50
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1082/676968946.py in <cell line: 0>()
      1 numbers = (10, 20, 30)
      2
----> 3 numbers[1] = 50

TypeError: 'tuple' object does not support item assignment
```

Next steps: Explain error

```
name = "Raj"
name[1] = "e"
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
/tmp/ipykernel_1082/975382128.py in <cell line: 0>()
      1 name = "Raj"
----> 2 name[1] = "e"

TypeError: 'str' object does not support item assignment
```

Next steps: Explain error

```
numbers = (5, 10, 15, 20)
```

```
total = 0

for num in numbers:
    if num % 10 == 0:
        total += num

print(total)
```

```
30
```

```
marks = (80, 90, 70, 100)

highest = marks[0]

for mark in marks:
    if mark > highest:
        highest = mark

print(highest)
```

```
100
```

```
fruits = ("Apple", "Banana", "Mango")

for i in range(len(fruits)):
    if i % 2 == 0:
        print(fruits[i])
```

```
Apple
Mango
```

```
numbers = (2, 4, 6)

result = 1

for num in numbers:
    result *= num

print(result + len(numbers))
```

```
51
```

```
numbers = (2, 4, 6)

result = 0

for num in numbers:
    result *= num

print(result + len(numbers))
```

```
3
```

```
student = ("Rahul", 85, 90, 95)

name = student[0]

average = sum(student[1:]) / 3

print(name, average)
```

```
Rahul 90.0
```

```
student[1:]
```

```
(85, 90, 95)
```

```
sum(student[1:])
```

```
270
```

```
sum(student[1:])/3
```

```
90.0
```

```
data = ("Python", "Java", "C++")

count = 0

for item in data:
    if len(item) > 4:
        count += 1

print(count)
```

```
1
```

HW

```
values = (5, 10, 15, 20)

total = 0

for value in values:
    if value % 5 == 0:
        total += value // 5

print(total)
```

```
colors = ("Red", "Blue", "Green")

for color in colors:
    print(color[0], end=" ")

print()
```

```
numbers = (1, 2, 3)

a, b, c = numbers

print(a + b * c)
```

```
marks = (80, 90, 95)

bonus = ()

for mark in marks:
    bonus += (mark + 5,)

print(bonus)
```

```
fruits = ("Apple", "Banana", "Mango")

for fruit in fruits:
    if "a" in fruit.lower():
        print(fruit)
```

```
numbers = (10, 20, 30, 40)

smallest = min(numbers)

largest = max(numbers)

print(largest - smallest)
```

```
prices = (100, 200, 150)

discount = ()

for p in prices:
    discount += (p - 20,)

print(sum(discount))
```

```
tuple1 = (1, 2)

tuple2 = (3, 4)

tuple3 = tuple1 + tuple2

print(tuple3 * 2)
```

```
data = (2, 4, 6, 8)

count = 0

for num in data:
    if num % 4 == 0:
        count += 1

print(count)
```